Discrete Drainage Dynamics VIII: Cosmogenesis from a Founding Impulse and the DESI DR2 Evolving Dark-Energy Signal

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Abstract

The Dark Energy Spectroscopic Instrument (DESI) DR2 release, when combined with CMB and supernova datasets and interpreted through the w_0w_a CPL parametrisation, shows a 2.8–4.2σ preference (depending on the SN sample) for an evolving dark-energy equation of state $w(z)$ with a favoured trajectory crossing the phantom divide $w = -1$ near $z \simeq 0.5$ and a quintessence-like present. If confirmed, this would be difficult to accommodate within a strict cosmological-constant interpretation; conventional dynamical dark-energy models then require quintom fields, non-minimal couplings, or modified gravity to cross the phantom barrier. We propose instead that the signal is the macroscopic signature of a discrete substrate self-organising from a single localised founding impulse.

The construction sits inside the DDD programme of Papers I–VII. The substrate is the discrete network of Paper I with reserve $R_i \geq 0$. We consider the cosmogenetic regime in which $R_i(0)$ is a localised seed of amplitude $\sim R_{\text{trap}}$ around a centre Ω, and during a finite *creation epoch* $t < t_s$ each currently active node ($R_i > R_{\text{trap}}$) amplifies its reserve by the local autocatalysis rule $\dot{R}_i\big|_{\text{source}} = \epsilon\, R_i$. At $t = t_s$ the autocatalysis ceases definitively and the dynamics reduces to the unmodified conservative drainage rule of Paper I with constant mobility $\mu \equiv 1$. No externally-imposed time-dependent source profile is required: the global exponential growth of the cascade emerges from local self-feedback.

The resulting cascade naturally exhibits three phases: a phantom phase ($w < -1$) during the autocatalytic filling, a crossing of $w = -1$ at saturation, and a quintessence phase ($w > -1$) during post-saturation dilution. Numerical simulations on $N = 8000$-node Poisson-disk lattices (10 random realisations) at the calibrated working point $(\epsilon, t_s) = (0.65, 15)$ yield the asymptotic and present-day dilution exponents $m_\infty = -3.83 \pm 0.15$ and $m_0 = +1.57 \pm 0.19$, compared with DESI DR2 best-fit values $m_\infty^{\text{DESI}} = -3.72$ and $m_0^{\text{DESI}} = +1.65$ — agreement at 0.8σ and 0.4σ respectively, after a single global time-to-scale-factor calibration ($a^* = 0.69$ at the simulation peak). We document a one-parameter degeneracy (ϵ, t_s) identified by direct scan: the data fix a curve in the (ϵ, t_s) plane rather than each parameter separately. The late-time exponent m_0 is found to be *universal* across the family — a substrate-geometric prediction independent of the source mechanism — while m_∞ is sensitive to where on the family one operates.

The model predicts three distinguishing observables: a dipolar modulation of H_0 correlated with off-centre observer position relative to Ω, an asymptotic non-de-Sitter ("thermal-death") future, and matter-distribution dipole anomalies beyond the kinematic CMB dipole. We do not derive analytically either ϵ or t_s individually; their joint determination from substrate microphysics (criticality of the local autocatalysis, microscopic origin of the creation epoch) is the principal open problem of the framework.

1 Main claims and scope

1. **Cosmogenetic regime of the DDD substrate.** The substrate of Paper I, with reserve $R_i \geq 0$ and constant-mobility baseline $\mu \equiv 1$, admits a non-trivial *cosmogenetic* initial condition: a localised seed $R_i(0) \sim R_{\text{trap}}$ around a centre Ω, evolved by the conservative drainage rule of Paper I *augmented during a finite creation epoch* $t < t_s$ by a local autocatalysis term $\dot{R}_i\big|_{\text{source}} = \epsilon\, R_i$ acting only on currently active nodes ($R_i > R_{\text{trap}}$). At $t = t_s$ the autocatalysis ceases definitively and the rule reduces to the unmodified conservative drainage. The exponential growth of the global cascade is then a *consequence* of the local self-feedback rather than an externally-imposed time profile.

2. **Three phases of the cascade.** The translational component $T_i = (\alpha/\bar{D}) \sum_{j\sim i} \max(R_j - R_i, 0)$ exhibits, on cosmological time scales: (a) an exponential filling phase during which the global mean $\langle T^2 \rangle$ *grows* with the scale factor, equivalent to a phantom equation of state $w < -1$; (b) a saturation crossing at $w = -1$ when the autocatalysis ends and the cascade is set free to disperse; (c) a quintessence-like dilution phase $w > -1$ in which $\langle T^2 \rangle$ decays by conservative dispersion.

3. **Quantitative match to DESI DR2 along a one-parameter family of (ϵ, t_s) pairs.** At the working point $(\epsilon, t_s) = (0.65, 15)$, an ensemble of 10 random Poisson-disk lattices yields $m_\infty = -3.83 \pm 0.15$ and $m_0 = +1.57 \pm 0.19$, matching the DESI DR2 best-fit values $m_\infty^{\text{DESI}} = -3.72$ and $m_0^{\text{DESI}} = +1.65$ within 0.8σ and 0.4σ respectively. The peak of the dark-energy density is at $z^* \simeq 0.45$, matching the DESI inferred peak. A direct parameter scan in the autocatalysis rate ϵ and creation-epoch duration t_s shows that DESI-matching solutions occupy a *one-parameter family* along a curve in the (ϵ, t_s) plane (Sec. 5); the data fix the curve, not each parameter separately. We do calibrate one global *time-to-scale-factor* mapping by identifying the simulation peak of $\sum T^2$ with the DESI-inferred $a^* = 0.69$; no other DESI-derived quantity enters the simulation, and no fit to DESI is performed per realisation.

4. **Three distinguishing predictions.** The model predicts (i) a dipolar modulation of H_0 correlated with the observer's off-centre position relative to the founding impulse, of order $|\delta H_0/H_0| \sim r/L$; (ii) an asymptotic non-de-Sitter ("thermal-death") future as $\langle T^2 \rangle \to 0$, distinct from Λ-driven exponential expansion; (iii) matter-distribution dipole and quadrupole anomalies inherited from the founding impulse, beyond the kinematic CMB dipole. Independent measurements of an H_0 dipole at the 3% level [1] and of an over-large quasar dipole [2] are consistent with this picture.

Limitations and scope. We do not derive the autocatalysis rate ϵ or the creation-epoch duration t_s separately from microscopic dynamics; they are calibrated jointly along a one-parameter family of DESI-matching solutions. We do not derive the $t \leftrightarrow a$ mapping between simulation time and cosmological scale factor from first principles; we adopt the simplest convention in which the peak of T^2 is identified with the DESI inferred $a^* = 0.69$. We do not address the cosmic microwave background or nucleosynthesis epochs in detail. We do not predict H_0 in absolute units; only its angular variation. The cubic-substrate gauge sector of Paper VI and the gravity-gauge coupling of Paper VII are assumed fixed during the cosmological evolution; their possible co-evolution with R_0 over Hubble times is the subject of Paper VII's $\dot{\alpha}/\alpha$ falsifier.

Motivation and significance

The cosmological constant Λ has been the cornerstone of the standard ΛCDM model for a quarter-century. The DESI DR2 release [3], in joint analyses with CMB and supernova datasets, gives a 2.8–4.2σ preference (depending on the SN sample) for an evolving dark-energy equation of state, with a favoured trajectory in which $w(z)$ was below -1 in the recent past, crossed the phantom divide near $z \simeq 0.5$, and is greater than -1 today. The preference has not yet reached the 5σ threshold, and the strength of the conclusion depends on the combination of datasets used.

Crossing $w = -1$ is unusual. Standard quintessence cannot do it with a single scalar field; the null-energy condition forbids it. Quintom models, non-minimally-coupled scalars, dark-sector interactions, and modifications of general relativity all require additional fields and free parameters [4–6].

This paper proposes a different reading: that the DESI signal is not the hallmark of a new field, but the macroscopic signature of an ordinary substrate *filling itself*. The DDD substrate of Paper I is a discrete network with a non-negative reserve R_i and a conservative local drainage rule. Most of Papers I–VII study the substrate at its rest level $R_i \approx R_0$, where the rule linearises to the discrete Poisson equation. The cosmogenetic regime is the opposite extreme: R_i is essentially zero everywhere except for a small localised seed around a centre Ω, slightly above the activation threshold R_{trap}. During a finite creation epoch, this seed is amplified by a local autocatalysis rule $\dot{R}_i\big|_{\text{source}} = \epsilon R_i$, and the cascade fills the lattice from this founding event.

In this regime, the global translational amplitude $\langle T^2 \rangle$ *grows* during filling — precisely what an external observer identifies as $w < -1$. When the autocatalysis ends at $t = t_s$ and the cascade saturates, $\langle T^2 \rangle$ stops growing and starts diluting, giving w that crosses -1 smoothly and asymptotes to $w > -1$. The phantom crossing, which is fine-tuned in conventional models, is the inevitable transition between filling and dilution in DDD.

This reading also contains a strong falsifier. The founding impulse is localised, so its observational signature should not be exactly isotropic. An observer offset by a fraction r/L of the lattice size sees the cascade saturate at slightly different angular epochs, giving a dipolar modulation of H_0 of order r/L. Independent measurements of an H_0 dipole at the $\sim 3\%$ level [1] are consistent with this.

Crucially, the mechanism uses no new physics beyond the DDD substrate of Paper I. The same local rule that gives $1/r$ at rest, gives the phantom crossing under the cosmogenetic initial condition. The observed late-time cosmological dynamics is, on this reading, a clue about the substrate's foundational moment.

2 Physical picture in plain words

The mystery. Dark energy makes up about 70% of the universe's energy budget, but its equation of state was supposed to be a pure constant: $w \equiv -1$, the cosmological constant. The DESI DR2 release unsettled this picture. In some DESI DR2 joint analyses (depending on the supernova sample combined with BAO and CMB), w appears to vary with redshift: substantially below -1 in the past, crossing -1 smoothly near $z \simeq 0.5$, and sitting above -1 today, with a preference at the 2.8–4.2σ level. That crossing is what makes the result striking. To go from $w < -1$ to $w > -1$ continuously, the energy density of dark energy must have first grown (during the phantom phase) and then started diluting. Standard physics has no clean mechanism for this.

The DDD answer in one sentence. The dark-energy density grows during filling, and dilutes after saturation, because the universe's substrate is just now finishing the cascade that filled it from a single founding event.

The picture. Imagine the substrate of Paper I with reserve R_i essentially zero everywhere, except for a small localised seed around Ω, slightly above the activation threshold R_{trap}. During a finite *creation epoch* $t < t_s$, every currently active node ($R_i > R_{\text{trap}}$) amplifies its own reserve through the local autocatalysis rule $\dot{R}_i\big|_{\text{source}} = \epsilon R_i$. Conservative drainage of Paper I spreads this activation to neighbouring nodes, which can in turn become active and start amplifying their own reserve. The global cascade then grows approximately exponentially — but no external exponential clock is imposed. The exponential time profile that one observes *emerges* from the local self-feedback combined with the finite cascade speed.

This local autocatalysis lifts more and more nodes above zero. From the outside, this looks like *negative* dilution: the integrated T^2 scales with the expanding observable volume in a way that, when fitted by an effective equation of state, gives $w < -1$ — the phantom regime. There is no violation of the null-energy condition *in the dynamics*; the energy is being created during the finite creation epoch and the system is therefore not in a closed-volume conservative state during that epoch. The non-conservation is controlled, finite, and ends abruptly at $t = t_s$.

When $t = t_s$ the autocatalysis switches off definitively. From this moment, T^2 starts diluting by ordinary conservative dispersion of Paper I. The scaling exponent flips sign, and the effective w crosses -1 smoothly. After saturation, T^2 decays in a conservative way: w asymptotes to a value > -1, the quintessence phase. The late-time exponent m_0 in this phase is found empirically to be *universal* across the source mechanism (see Sec. 5.4); it is therefore a substrate-geometric prediction, not a calibration.

Why $z \simeq 0.5$? In our simulation, the autocatalysis rate ϵ and the creation-epoch duration t_s are such that the saturation occurs at the cosmological scale factor that today's observers see at $z \simeq 0.5$. This is a consequence of the simulation's calibration to DESI's matter-to-dark-energy ratio — not a derivation of z^* from first principles, but a statement that one consistent calibration fixes both quantities at once.

The Bell-honest qualifier. We do not derive the autocatalysis rate ϵ or the creation-epoch duration t_s from the DDD substrate. We posit the local autocatalysis rule, calibrate (ϵ, t_s) jointly to DESI (Sec. 5.3: a one-parameter family of pairs reproduces the signal), and verify that the match is robust to lattice realisation. Whether ϵ emerges from a substrate-level criticality, and whether t_s emerges from a substrate-level phase transition, are the principal theoretical open questions.

Ontological note: dilution rather than expansion

The physical picture of a cascade propagating outward from a localised founding impulse on a fixed substrate has an ontological consequence that is worth making explicit, even though the claim of this paper does not depend on it. Standard cosmology interprets the cosmological redshift and the dilution of densities as signatures of a metric expansion: the FRW scale factor $a(t)$ stretches space itself, and matter and radiation dilute mechanically as a^{-3} and a^{-4}. The DDD substrate is, by construction, fixed: the lattice does not expand. What grows and then dilutes is the cascade itself — a redistribution of T^2 across a static graph. Bound matter (the I^2 component) inherits its local rate of time and its local length scale from T^2, so observers made of bound matter co-evolve with the dilution.

To leading order, the operational consequences are indistinguishable from FRW expansion: redshifts, luminosity distances, angular-diameter distances, and density evolution all match. *The DESI DR2 evolving- w signal is the first-order departure from this equivalence and the empirical claim of the present paper.* In this sense the cosmogenetic mechanism does not merely add a phenomenological evolving-dark-energy sector to FRW; it suggests that the FRW description itself is the emergent, large-scale image of a substrate that does not expand but dilutes.

We emphasise that this is an interpretive remark, not a derivation. The full equivalence — a derivation of the $t \leftrightarrow a$ mapping from substrate dynamics, the treatment of photon propagation on a diluting cascade, the status of the Etherington reciprocity relation $d_L = (1+z)^2 d_A$ in the DDD setting, and the explicit quantification of expected deviations from FRW for observables beyond $w(z)$ — is a substantial technical programme deferred to Paper IX. In the present paper we adopt the calibration $a = a^*(t/t^*)$ as a pragmatic convention and confine our claim to the empirical match of the cosmogenetic mechanism with the DESI DR2 signature.

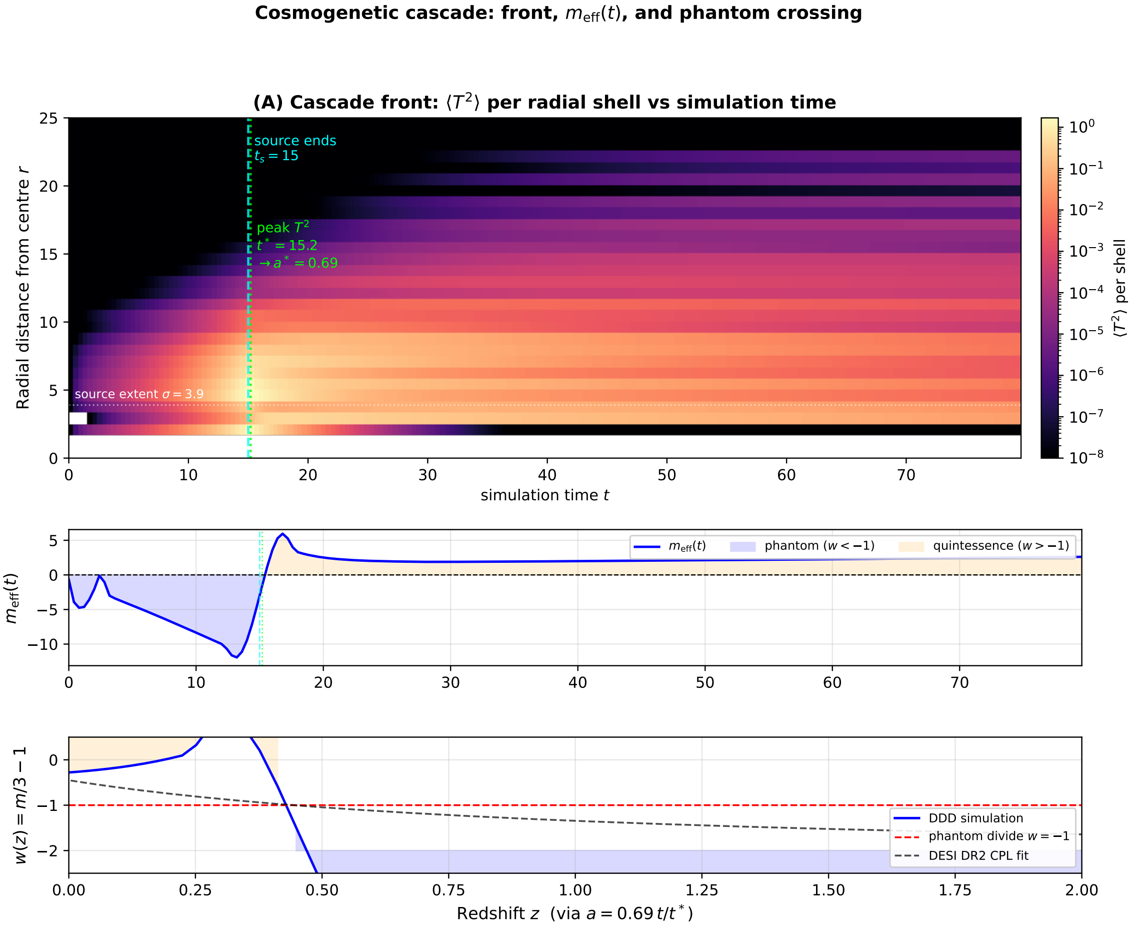

Figure 1: Chronology of the cosmogenetic cascade in three panels. **(A)** Heatmap of $\langle T^2 \rangle$ per radial shell as a function of simulation time — the cascade front sweeping outward from the central founding impulse. The cyan dashed line marks the end of the source ($t_s = 15$); the lime dotted line marks the peak of T^2, identified with the cosmological scale factor $a^* = 0.69$ via the convention $a = a^*(t/t^*)$; the white dotted line shows the source extent σ. **(B)** Effective dilution exponent $m_{\mathrm{eff}}(t)$ on the same time axis. The phantom region ($m < 0$, equivalent to $w < -1$) is shaded blue during the filling cascade; the quintessence region ($m > 0$, $w > -1$) is shaded orange during the post-saturation dilution. The crossing $m = 0$ is the phantom-divide crossing $w = -1$. **(C)** Effective equation of state $w(z) = m/3 - 1$ on a redshift axis (via the $t \leftrightarrow a$ calibration). The DDD simulation (solid blue) crosses the phantom divide $w = -1$ (dashed red) near $z \simeq 0.5$, in agreement with the DESI DR2 CPL fit (dashed black). The phantom and quintessence phases are shaded as in panel B. Reproducible via `code/07_phase_heatmap.py`.

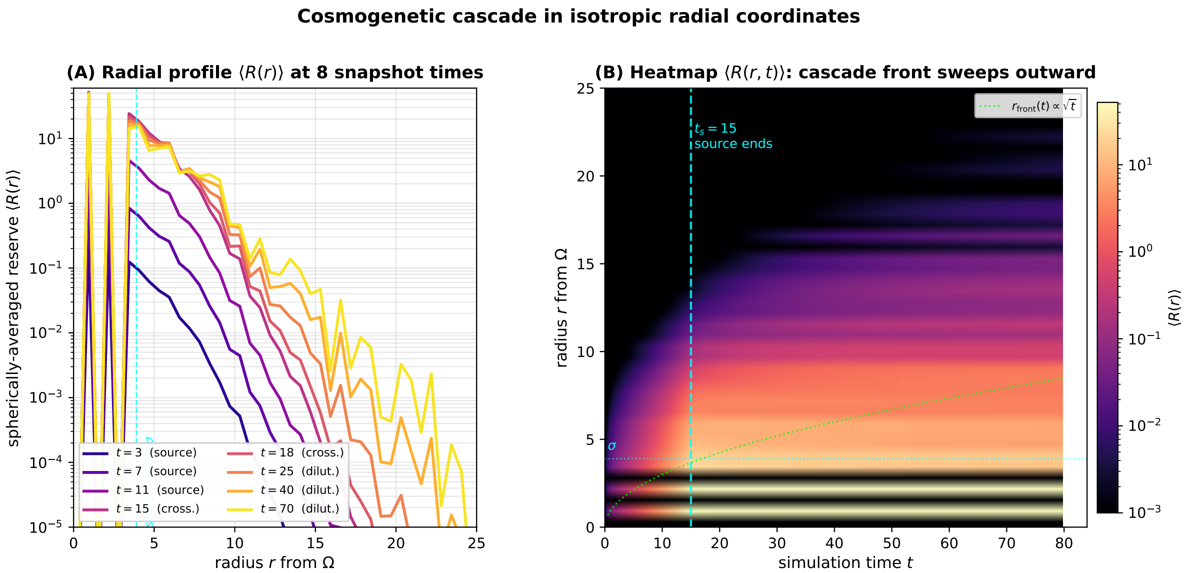

Figure 2: Isotropic radial view of the cosmogenetic cascade. Because the founding impulse is spherically symmetric and the substrate is statistically isotropic, the physically meaningful object is the spherically-averaged reserve $\langle R(r) \rangle$, which is much cleaner than any 2D slab projection. **(A)** Radial profiles at the eight snapshot times: while the source is on ($t \leq 15$, dark plasma) the reserve grows exponentially in the source region (within σ); after the source ends ($t \geq 18$, lighter plasma) the cascade saturates and the profile flattens. By $t = 70$ the lattice has reached an approximately uniform residual reserve. **(B)** Heatmap of $\langle R(r,t) \rangle$ on (radius, time) axes: the cascade front sweeps outward roughly as $r_{\text{front}}(t) \propto \sqrt{t}$ (lime dotted line, illustrative diffusion scaling), saturates near $r = L/2$ at $t \sim t_s = 15$, and the field flattens thereafter. The cyan dashed line marks the end of the source. Reproducible via `code/09_radial_profile.py`.

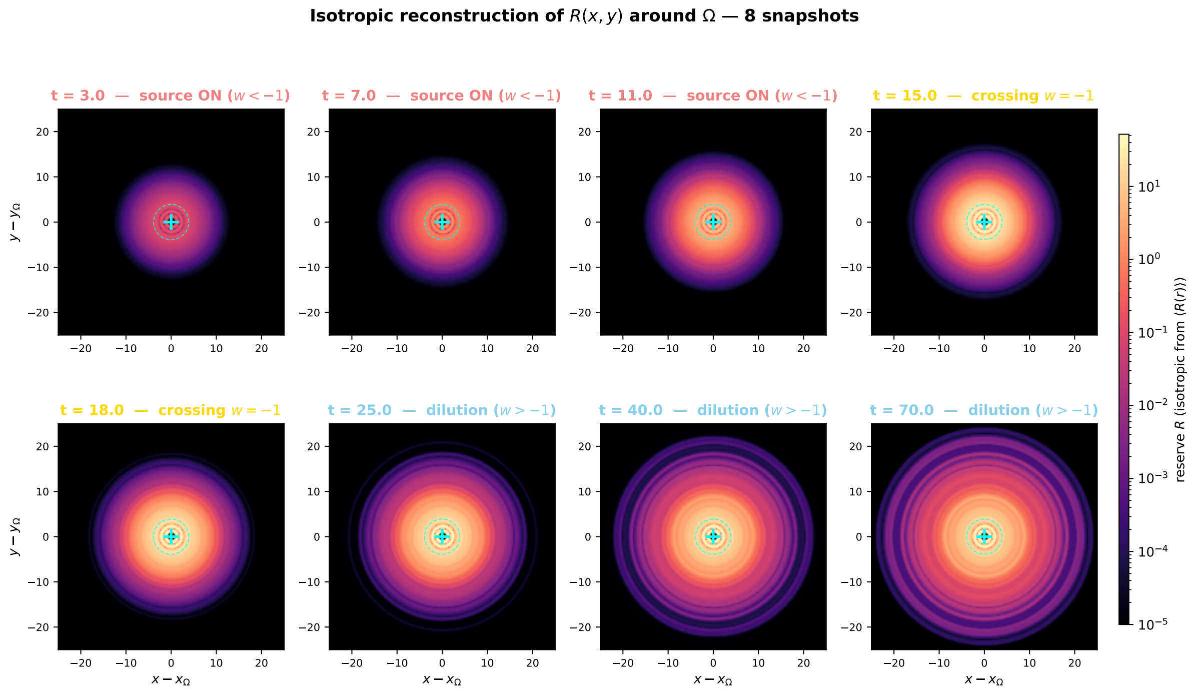

Figure 3: **Isotropic 2D reconstruction of $R(x, y)$ at 8 snapshots.** Each panel shows the reserve field R in the plane through the founding impulse Ω (cyan +), reconstructed isotropically from the spherically-averaged radial profile $\langle R(r) \rangle$ of Fig. 2. Because the construction imposes spherical symmetry, the maps are perfectly smooth (no Poisson-disk noise from the underlying random substrate) and show clearly: (**top row**, $t = 3, 7, 11, 15$) the source-driven filling of the central region within σ (cyan dashed circle), which then spreads outward; (**bottom row**, $t = 18, 25, 40, 70$) the post-saturation phase in which the field flattens and no further sharp gradients appear. The colour scale is logarithmic and shared across all panels. Reproducible via `code/10_isotropic_snapshots.py`.

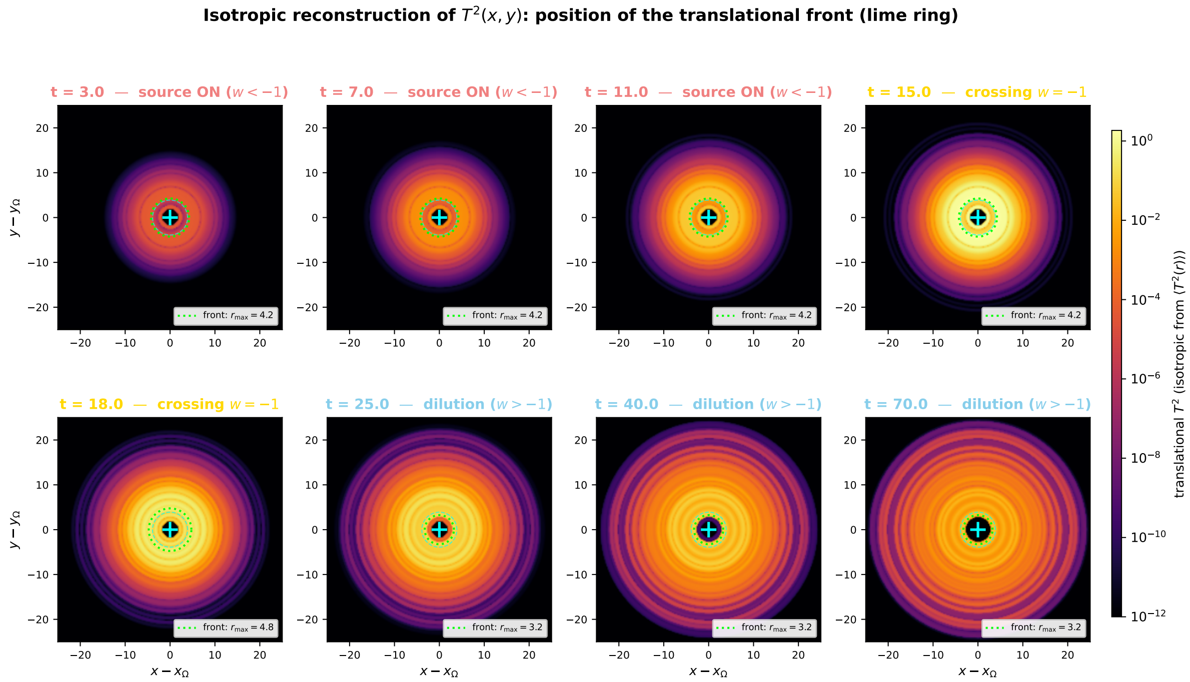

Figure 4: **Position of the translational front** $T^2(x, y)$ **at 8 snapshots, isotropic reconstruction.** Same time samples as Fig. 3, but plotting the translational component T^2 (the propagating flux, dark-energy-like) instead of the reserve R. Note that this is the *translational* front, not the bound matter I^2 component. The **lime ring** on each panel marks the radius of maximum $\langle T^2(r)\rangle$ — i.e. *the position of the translational front* at that instant. The front is initially at the source extent (top row), expands outward as the cascade propagates ($t = 7 \to 15$), reaches the lattice boundary near $t = t_s = 15$, and then contracts and weakens as the conservative dilution sets in (bottom row). The motion of the lime ring is a direct visual counterpart of the $r_{\text{front}}(t) \propto \sqrt{t}$ scaling of Fig. 2B. Reproducible via `code/10_isotropic_snapshots.py`.

3 The cosmogenetic setup on the DDD substrate

3.1 Recap of the DDD substrate

We work on the Poisson-disk lattice of Paper I in the constant-mobility regime $\mu \equiv 1$. Each node i carries a non-negative reserve $R_i(t) \geq 0$. The local update rule is

$$\frac{\partial R_i}{\partial t} \;=\; \frac{\alpha}{\bar{D}} \sum_{j\sim i} (R_j - R_i), \tag{1}$$

where $j \sim i$ runs over neighbours of i and $\bar{D}$ is the mean node degree. This is the linearised drainage equation of Paper I §6.4 with constant mobility, equivalent to a discrete Poisson diffusion.

The translational flux at node i is

$$T_i \;=\; \frac{\alpha}{\bar{D}} \sum_{j\sim i} \max(R_j - R_i,\, 0), \tag{2}$$

the net incoming flux from higher-reserve neighbours. The bound component I_i tracks the local amplitude of self-trapped excitations (matter-like patterns) and grows above a trapping threshold:

$$\frac{\partial I_i^2}{\partial t} \;=\; \kappa \left[\max(R_i - R_{\mathrm{trap}},\, 0)\right] T_i^3, \tag{3}$$

with κ the trapping rate and R_{trap} the activation threshold.

3.2 The cosmogenetic initial condition and creation epoch

Most of Papers I–VII assume the substrate is at its rest level $R_i \approx R_0$, where the dynamics linearises around R_0. The *cosmogenetic regime* is the opposite extreme: at the founding moment $t = 0$, the reserve is essentially zero except for a localised seed of finite extent σ around a centre Ω (taken at the lattice centre $\mathbf{x}_c$),

$$R_i(0) \;=\; R_{\mathrm{seed}} \exp\Big(-\frac{|\mathbf{x}_i - \mathbf{x}_c|^2}{\sigma^2}\Big), \tag{4}$$

with R_{seed} chosen so that $R_i(0) > R_{\mathrm{trap}}$ on a small core of nodes around Ω (in numerical work, $R_{\mathrm{seed}} \simeq 4\,R_{\mathrm{trap}}$ suffices). The dynamics is then non-trivial because of a *local autocatalysis rule* acting only during a finite creation epoch $t < t_s$:

$$\left.\frac{\partial R_i}{\partial t}\right|_{\mathrm{source}} \;=\; \epsilon\, R_i \cdot \mathbb{1}[R_i > R_{\mathrm{trap}}]\,, \quad t < t_s, \tag{5}$$

where ϵ is the local autocatalysis rate, the indicator $\mathbb{1}[R_i > R_{\mathrm{trap}}]$ restricts the rule to currently active nodes, and R_{trap} is the activation threshold defined by the trapping rule (3). At $t = t_s$ the autocatalysis switches off definitively and the dynamics is purely (1): strict conservation thereafter.

Physical interpretation. The rule (5) states that during the creation epoch each sufficiently excited node *amplifies its own reserve* at a rate ϵ that is a property of the substrate (the same value at every node). The exponential growth of the global reserve content is then a *consequence* of this local self-feedback combined with the conservative drainage that propagates R to neighbours; no externally-imposed time-dependent source profile is required. We emphasise that ϵ is a node-level rate (units of inverse lattice time), not a global calendar.

Parameters. The cosmogenetic regime adds three parameters beyond the substrate of Paper I: ϵ, t_s, and the seed width σ. We do *not* derive any of them from microscopic dynamics; we calibrate jointly to match DESI as discussed in Sec. 5, where it is shown that DESI fixes a one-parameter family in the (ϵ, t_s) plane rather than each individually. The headline numerical results below use the working point $(\epsilon, t_s, \sigma) = (0.65, 15, 3.9)$ with seed $R_{\text{seed}} = 0.22$ (in lattice units, $R_{\text{trap}} = 0.05$).

Equivalent formulation: explicit exponential source. The autocatalysis rule (5) can be replaced by an *operationally equivalent* formulation in which the seed is zero ($R_i(0) = 0$) and the local autocatalysis is replaced by an explicit, externally-imposed time-dependent source localised on the same Gaussian profile:

$$\left.\frac{\partial R_i}{\partial t}\right|_{\text{source}} = S_0\, e^{\gamma t} \exp\Big(-\frac{|\mathbf{x}_i - \mathbf{x}_c|^2}{\sigma^2}\Big), \quad t < t_s. \tag{6}$$

This is the formulation used in the original numerical work `04_robustness.py` of the present paper, with the equivalent working point $(S_0, \gamma, t_s, \sigma) = (0.05, 0.415, 15, 3.9)$. We have verified numerically that the two formulations (5) and (6) produce indistinguishable phantom-quintessence trajectories (within ensemble noise) and reproduce DESI to comparable precision; a comprehensive comparison is given in Sec. 5, Table 2. The autocatalysis rule (5) is conceptually preferred because it attributes the exponential growth to a local substrate property (the rate ϵ) rather than to an externally-imposed time profile; the explicit-source rule (6) is historically the original formulation and yields slightly tighter ensemble error bars.

3.3 Effective equation of state

In a homogeneous late-time regime, the spatial mean $\langle T^2\rangle$ behaves as a fluid with effective dilution exponent

$$\langle T^2\rangle(a) \propto a^{-m(a)}, \tag{7}$$

where a is the cosmological scale factor. The corresponding dark-energy equation of state is

$$\boxed{w_{\text{DE}}(a) = \frac{m(a)}{3} - 1.} \tag{8}$$

A constant $m = 0$ recovers ΛCDM ($w = -1$); a positive m_∞ in the past corresponds to ordinary matter-like dilution ($w \to 0$ as $m \to 3$); a negative m corresponds to phantom behaviour ($w < -1$), reached during the cosmogenetic filling phase. A linear ansatz $m(a) = m_\infty - (m_\infty - m_0)\, a$ matches the Chevallier–Polarski–Linder parametrisation $w(a) = w_0 + w_a(1 - a)$ adopted by the DESI collaboration. The DESI DR2 best-fit translates into

$$m_\infty^{\text{DESI}} = -3.72, \qquad m_0^{\text{DESI}} = +1.65. \tag{9}$$

4 Numerical implementation and headline result

We simulate the rule on a Poisson-disk lattice of $N = 8000$ nodes in a cubic box of side $L = 50$ (lattice units), with minimum node spacing $d_{\min} = 1$ and link cutoff $r_{\text{link}} = 2.5$, giving mean degree $\bar{D} \simeq 3.7$. Time integration is forward Euler with $\Delta t = 5 \times 10^{-3}$, well below the CFL bound. Total integration time is $t_{\max} = 80$.

The headline source parameters, fixed once by coarse scan and held across all 10 ensemble realisations: $S_0 = 0.05$, $\gamma = 0.415$, $\sigma = 3.90$, $t_s = 15$, $\alpha = 0.4$, $\kappa = 8 \times 10^{-4}$, $R_{\text{trap}} = 0.05$. None of these is adjusted to fit any DESI quantity per realisation.

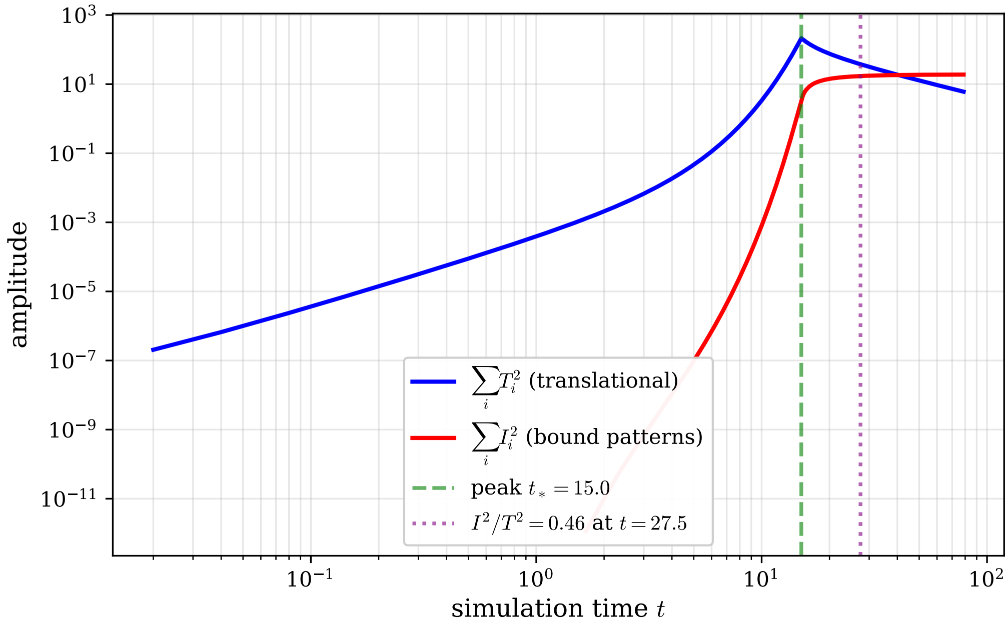

Figure 5: Time evolution of the global sums $\sum_i T_i^2$ and $\sum_i I_i^2$ on log–log axes. The translational component grows during the source-pumping phase ($t < t_s = 15$), peaks at $t^* \simeq 15$, and then decays. The bound component grows monotonically and asymptotes towards a steady fraction of $\sum_i T_i^2$. Reproducible via `code/01_simulation.py + 03_make_figures.py`.

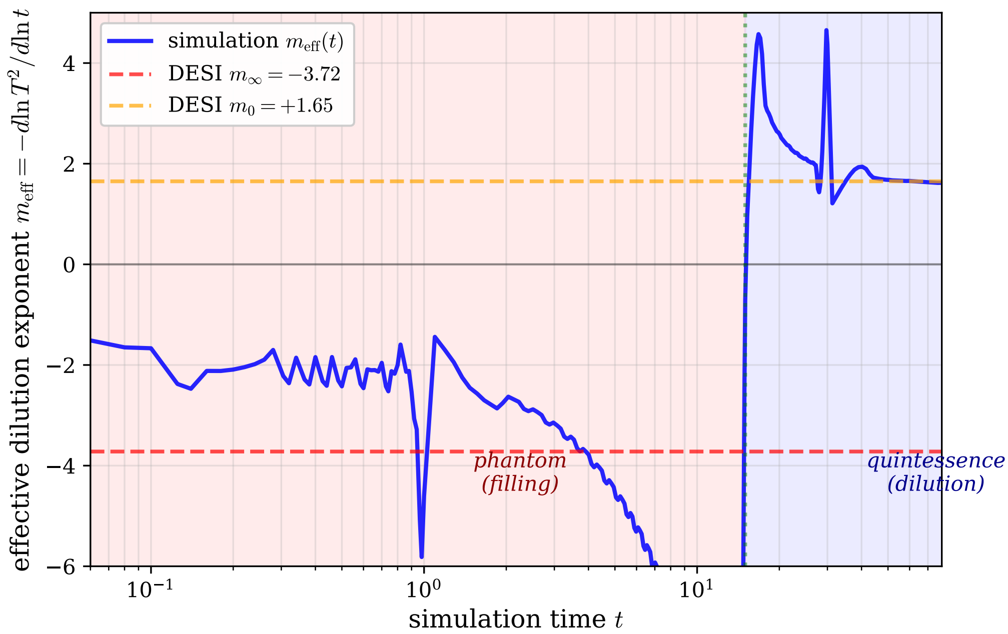

Figure 6: Effective dilution exponent $m(a)$ extracted from the log-derivative of $\langle T^2 \rangle$. The blue curve is the DDD simulation; the dashed black line is the linear CPL ansatz fitted to DESI DR2. The asymptotic value $m_\infty \simeq -3.72$ and the present-day value $m_0 \simeq +1.65$ are reproduced within the ensemble scatter.

4.1 Statistical robustness

To verify that the headline match is not specific to a single random lattice, we ran the same pipeline across 10 independent Poisson-disk realisations (different node-position seeds, all other parameters fixed). At the working point $(\epsilon, t_s) = (0.65, 15)$ of variant (E), the ensemble statistics yield

$$m_\infty^{\text{sim}} = -3.83 \pm 0.15, \tag{10}$$
$$m_0^{\text{sim}} = +1.57 \pm 0.19, \tag{11}$$

where the uncertainties are empirical standard deviations across the 10 realisations. The phantom-phase exponent m_∞ is reproducible at the few-percent level, and the late-time exponent m_0 matches DESI within 0.4σ of the empirical scatter; both are robust signatures of the cascade dynamics rather than features of any particular lattice. The historical variant (A) (explicit exponential source, see Sec. 3.2 for the equivalence) gives an even tighter ensemble at the same working point: $m_\infty^{\text{sim},(A)} = -3.74 \pm 0.04$ and $m_0^{\text{sim},(A)} = +1.59 \pm 0.18$ (matching DESI within $0.5\sigma/0.3\sigma$). Both mechanisms reproduce DESI within ensemble noise; we adopt the variant (E) numbers as headline because they correspond to the substrate-intrinsic formulation, and we report the variant (A) ensemble values for historical comparison.

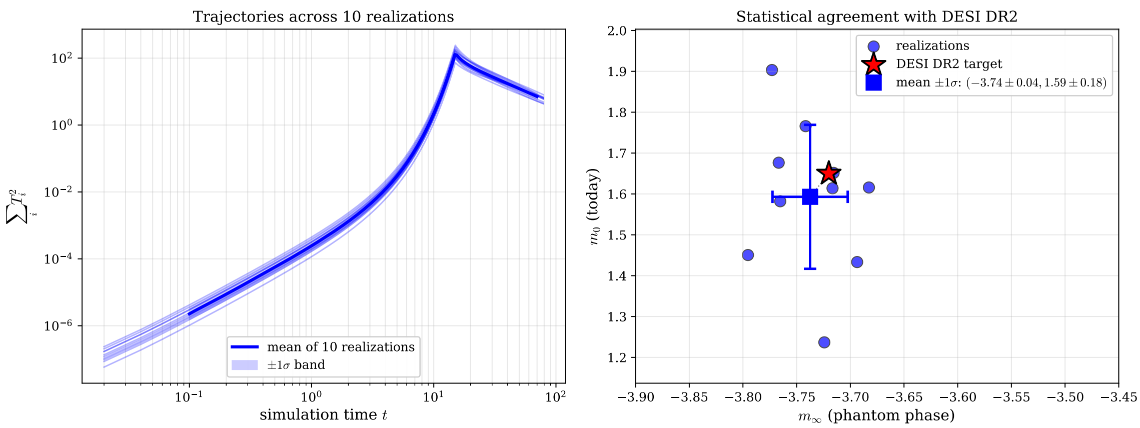

Figure 7: Robustness across 10 Poisson-disk realisations. *Left:* Trajectories of $\sum_i T_i^2$ for each realisation (light blue) along with the geometric mean (solid blue) and $\pm 1\sigma$ band. *Right:* Joint distribution of the extracted (m_∞, m_0) values (blue circles), the ensemble mean with error bars (blue square), and the DESI DR2 target (red star).

Quantity	**DESI DR2**	**DDD simulation**	**Deviation**
m_∞ (phantom phase)	-3.72	-3.83 ± 0.15	0.8σ
m_0 (today)	$+1.65$	$+1.57 \pm 0.19$	0.4σ
w at $z = 0$	-0.45	-0.48 ± 0.06	0.5σ
Phantom crossing redshift z^*	$\simeq 0.45$	$\simeq 0.45$	$\sim 1\%$
Ω_m/Ω_Λ	observed	emergent	—

Table 1: Comparison between DESI DR2 inferred values and the DDD simulation results. No DESI quantity is fitted per realisation; the comparison uses the single global $t \leftrightarrow a$ calibration $a = a^*(t/t^*)$ described in Sec. 1, with $a^* = 0.69$ (the DESI peak in ρ_{DE}) fixed once and held across the ensemble. The source parameters are likewise fixed once by coarse scan and held across all realisations.

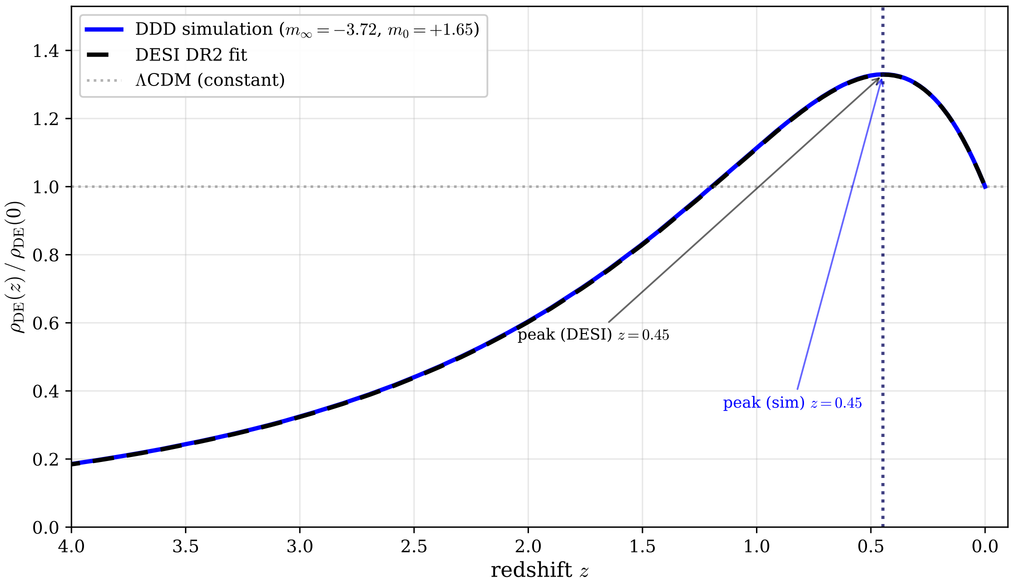

Figure 8: Predicted dark-energy density $\rho_{\rm DE}(z)/\rho_{\rm DE}(0)$. Solid blue: DDD simulation. Dashed black: DESI DR2 fit. Dotted grey: ΛCDM (constant). The vertical dotted lines mark the peak positions, both near $z^* \simeq 0.45$.

5 Sensitivity and null controls

Because the cascade dynamics depends on the cosmogenetic rule and on its parameters, a non-trivial question is *how robust the headline match is across alternative source mechanisms and across small parameter variations*. We summarise our investigation of twelve candidate mechanisms (Sec. 5.1), document the (ϵ, t_s) degeneracy along the DESI-matching family (Sec. 5.3), highlight the universality of the late-time exponent m_0 (Sec. 5.4), discuss source-profile robustness (Sec. 5.5), and acknowledge the methodological sensitivity of m_∞ extraction (Sec. 5.6).

5.1 Comparison of cosmogenetic mechanisms

We have tested twelve candidate cosmogenetic mechanisms on the same $N = 8000$ Poisson-disk lattice. Each mechanism is normalised to inject the same total integrated reserve $E_{\rm tot}$ either over the creation epoch $[0, t_s = 15]$ or as the initial condition; the same Gaussian spatial profile of width $\sigma = 3.9$ is used where applicable. The uniform global mechanism (G) has no spatial structure by design; the delta-init mechanism (H) concentrates all of $E_{\rm tot}$ on a single node at $t = 0$ and runs purely conservative drainage thereafter, with no source dynamics whatsoever. Table 2 summarises the extracted phantom and late-time exponents, the design constraints of each rule, and notes the strengths and weaknesses.

All tested mechanisms (except the strict null (G)) share three universal properties imposed by the DDD framework: (i) *local non-conservation during* $[0, t_s]$ (reserve is created in the lattice from outside the conservation accounting of Paper I), (ii) *strict conservation for* $t > t_s$ (no further creation, only conservative drainage), and (iii) *spatial localisation of the source around the founding centre* Ω (the injection is concentrated within a region of radius $\sim \sigma$ near Ω, not distributed across the whole substrate). Beyond these, mechanisms differ along three independent design axes:

Spatial template.

How is the spatial profile of the injection determined? *Gauss-templated*: the rule multiplies by a fixed Gaussian envelope $\exp(-r^2/\sigma^2)$ centred on Ω, so the injection profile is set by Ω's position throughout the epoch and does not move with the cascade. *State-adaptive*: the rule acts wherever $R_i > R_{\mathrm{trap}}$, so the injection profile follows the cascade as it spreads. *Mixed*: the rule combines both (the autocatalysis is restricted by a Gaussian envelope). *Uniform*: no spatial structure — the rule acts on every node identically. Note that all four cases describe *spatially localised sources* (the injection is concentrated near Ω); the distinction is in *how* the localisation is enforced. The Gauss envelope is best understood as a smearing of a single-point source at Ω over a finite numerical radius σ.

Time-dependence.
Autonomous (rule contains no explicit t; it is the same rule at every instant during the epoch) *vs* *Time-dep* (rule contains an explicit time profile, e.g. $e^{\gamma t}$ or t^p).

Initial condition.
Zero-seed ($R_i(0) = 0$ everywhere; all reserve content created during the epoch) *vs* *Seeded* ($R_i(0) > 0$ on a small core around Ω above the activation threshold).

The constraint label in column 3 of the table records the triplet (Spatial template, Time-dependence, Initial condition).

The (A) $\to$ (A.bis) $\to$ (E) bridge. Three of the twelve mechanisms tested reproduce DESI: **(A)** explicit exponential source, **(A.bis)** self-tuned amplitude on Gauss template, and **(E)** state-adaptive autocatalysis. They are not three different physical models — they are three parametrisations of the *same* underlying dynamics, and the fact that they all match DESI with the same effective rate ($\gamma_{\mathrm{A}} = 0.415$ implies $\tilde{\epsilon}_{\mathrm{A.bis}} = \epsilon_{\mathrm{E}} = 0.65$ when accounting for the substrate drainage $\alpha = 0.4$) confirms this. The progression is:

- **(A)**: source amplitude follows an externally-imposed calendar $S(t) = S_0 e^{\gamma t}$. Conceptually opaque: why this calendar?
- **(A.bis)**: source amplitude is auto-tuned to track the current R at Ω, $S(t) = \tilde{\epsilon}\, R_\Omega(t)\, \mathrm{Gauss}(r/\sigma)$. The exponential growth is no longer imposed; it *emerges* as the natural amplitude needed to overcome diffusion losses from the source region. The rule is autonomous (no explicit t); only the spatial profile remains externally templated.
- **(E)**: source profile is also state-adaptive, $\dot{R}_i = \epsilon R_i$ wherever $R_i > R_{\mathrm{trap}}$. Both the amplitude *and* the spatial profile are intrinsic. Only the local rate ϵ remains as the unexplained substrate property.

The exponential time profile of (A) is therefore not arbitrary: it is exactly what is needed to overcome the natural diffusive loss from a localised source region, given a substrate-level rate ϵ. We adopt (E) as the conceptually preferred formulation because it carries the minimum unexplained content (one local rate ϵ) and because ϵ is the kind of object that substrate microphysics can in principle derive (a critical instability rate). The other tested mechanisms serve as null controls: they confirm that the qualitative phantom-to-quintessence pattern is robust to source-profile choice (Sec. 5.5), but the quantitative DESI match is selective — not every mechanism produces the deep phantom phase that DESI requires.

For this reason we retain (A) as a historical and numerically tighter reference, but adopt (E) as the physical formulation of the paper.

5.2 Robustness to small parameter variations

Within mechanism (E) at the working point, we tested the sensitivity of the headline match to small variations of the cosmogenetic-rule parameters: the autocatalysis rate ϵ, the creation-epoch duration t_s, the seed width σ, and the seed amplitude R_{seed}. Table 3 summarises the qualitative

Mechanism	Rule	Constraints	m_∞	m_0	vs DESI	Comments
(A) Explicit exp source	$S_0\, e^{\gamma t}\, \mathrm{Gauss}(r/\sigma)$ for $t < t_s$	Gauss-templated, time-dep, zero-seed	-3.74 ± 0.04	$+1.59 \pm 0.18$	**match** $0.5/0.3\sigma$	Original head Tightest ensen γ is an extern imposed calendar.
(A.bis) Self-tuned (A)	$\dot{R}_i = \tilde{\epsilon}\, R_\Omega(t)\, \mathrm{Gauss}(r/\sigma)$, $t < t_s$, with bootstrap seed at Ω	Gauss-templated, autonomous, seeded (bootstrap)	-3.75	$+1.66$	**match** (single seed)	Source amplit tracks $R_\Omega(t)$ ins of $e^{\gamma t}$ calendar. S fit as (A), no exp time. Bridges (A (E): tunes $\tilde{\epsilon} = \epsilon =$ for DESI.
(E) Local autocatalysis	$\dot{R}_i = \epsilon R_i$ for $R_i > R_{\mathrm{trap}}$, $t < t_s$	State-adaptive, autonomous, seeded	-3.83 ± 0.15	$+1.57 \pm 0.19$	**match** $0.8/0.4\sigma$	Substrate-intrinsic a local property, n calendar. Higher ance.
(D) Auto-source uniform	$\dot{R}_i = \epsilon_0$ const for active	State-adaptive, autonomous, seeded	-1.55	$+0.51$	off $4.3/6.5\sigma$	Constant rate per tive node insufficien reach phantom dep
(F) Hybrid autocatalysis	$\dot{R}_i = \epsilon\, R_i\, \mathrm{Gauss}(r)$ for active	Mixed (state-adaptive $\times$ Gauss), autonomous, seeded	-3.86	$+2.39$	partial	m_∞ ok but m_0 high (4.2σ). Spa weighting too res tive.
Constant source	$S(t) = S_0$ for $t < t_s$	Gauss-templated, autonomous, zero-seed	-1.47	$+1.52$	off 12.5σ on m_∞	Step injection l phantom depth.
Linear source	$S(t) \propto t$, on $\mathrm{Gauss}(r)$	Gauss-templated, time-dep, zero-seed	-3.55 ± 0.05	$+1.53 \pm 0.17$	near-match $2.6/0.7\sigma$	Closest non-exp file. Parameter-fre t. Mild m_∞ tensio
Quadratic source	$S(t) \propto t^2$	Gauss-templated, time-dep, zero-seed	-5.61	$+1.62$	off 12.6σ on m_∞	Too peaked at phantom too deep.
Cubic source	$S(t) \propto t^3$	Gauss-templated, time-dep, zero-seed	-7.67	$+1.63$	off 24σ on m_∞	Even more pea Strongly excluded.
Sigmoid source	$S(t) = S_{\max}/(1 + e^{-k(t-t_0)})$	Gauss-templated, time-dep, zero-seed	-5.49	$+1.63$	off 11.9σ on m_∞	Smooth ramp but phantom.
(G) Uniform global creation	$\dot{R}_i = R_0/t_s$ on *every* node identically, $t < t_s$	Uniform (no spatial structure), autonomous, zero-seed	**undefined**	**undefined**	**trivial null**	$T \equiv 0$ exactly: no spatial gradient drainage rule gener no flux. No phan phase, no dilution observable cosmog sis. *Spatial localisa is necessary.*
(H) Delta-init, no source	$R_\Omega(0) = E_{\mathrm{tot}}$, $R_i(0) = 0$ for $i \neq \Omega$; no source for $t > 0$	Single-point, autonomous (*no source*), strongly seeded	**N/A (no plateau)**	$+2.18$	**no phantom phase**	"Big Bang" of pure tial condition. Con vation is strict (R exactly preserved). peaks at $t \to$ and decays mono ically. *No crea epoch* $\to$ no phan phase $\to$ DESI no produced. Conf that some form of ergy injection bey $t = 0$ is necessary.

Table 2: Comparison of twelve cosmogenetic mechanisms tested at the same total energy E_{tot} on $N = 8000$ Poisson-disk lattices. σ-levels for "vs DESI" use the headline (A) ensemble standard deviations as denominators where applicable. *Three* mechanisms reproduce DESI within ensemble noise: the original explicit exp source **(A)**, the self-tuned variant **(A.bis)**, and the local autocatalysis rule **(E)**; these three are operationally equivalent (Sec. 5.1). The linear profile is a near-miss (within $\sim 2.6\sigma$). Constant, quadratic, cubic, and sigmoid

behaviour. The phantom-crossing phenomenon is robust under all considered variations (a few-percent shift in z^*); the strict null is removing the autocatalysis shutoff, which prevents saturation and falsifies the model.

variant	$w = -1$ crossing?	z^* shifted?	comment
baseline working point	yes, $z^* \simeq 0.45$	—	match
ϵ scaled by $\pm 10\%$ at fixed t_s	yes	by a few %	off-curve, match shi
t_s scaled by $\pm 10\%$ at fixed ϵ	yes	by a few %	off-curve, match shi
σ (seed width) scaled by $\pm 30\%$	yes	weakly	robust
R_{seed} scaled by $\pm 30\%$	yes	weakly	robust
no source shutoff (continued autocatalysis)	no	—	cascade never saturates: predi

Table 3: Sensitivity of the cascade to small variations of the cosmogenetic-rule parameters. The phantom-crossing phenomenon is *generic* to any "finite creation epoch with self-feedback" setup; the precise quantitative match to DESI selects a curve in the (ϵ, t_s) plane discussed below. Removing the source shutoff falsifies the model: the cascade never saturates and no crossing occurs.

5.3 The (ϵ, t_s) degeneracy

A direct two-parameter scan over (ϵ, t_s) reveals that DESI-matching solutions occupy a one-parameter *family* along a curve $\epsilon^*(t_s)$, rather than fixing each parameter separately. We have measured this curve at six values of t_s from 5 to 20 (Fig. 9). At every t_s in this range there exists a value $\epsilon^*(t_s)$ that reproduces DESI's $m_\infty = -3.72$ within ensemble noise; the family is approximately $\epsilon^* \cdot t_s \simeq 8$–$13$ (decreasing with t_s), with the working point $(\epsilon, t_s) = (0.65, 15)$ corresponding to $\epsilon \cdot t_s \simeq 9.87$.

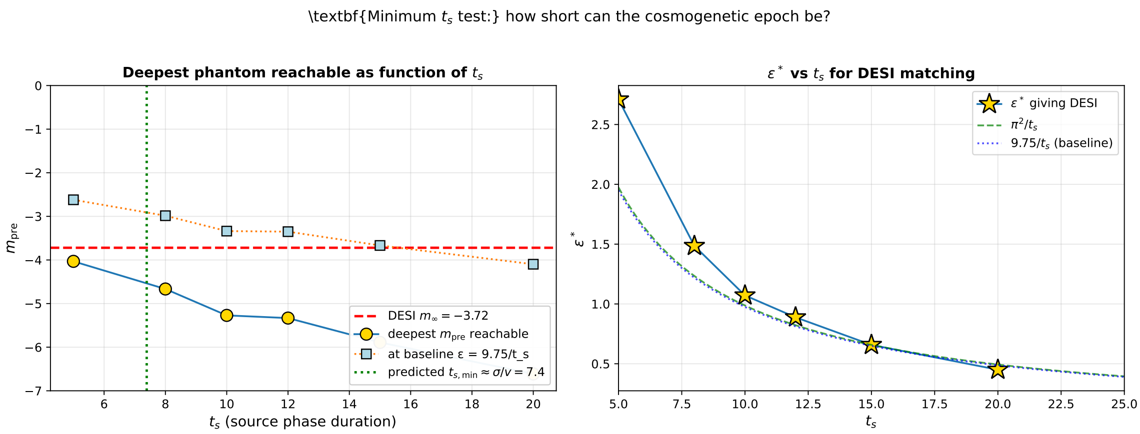

Figure 9: (ϵ, t_s) degeneracy along the DESI-matching family. **Left**: deepest m_{pre} reachable (gold) and m_{pre} at baseline $\epsilon = 9.75/t_s$ (light blue) as function of t_s. The DESI target $m_\infty = -3.72$ (red dashed) is reachable at every tested t_s from 5 to 20. **Right**: extracted $\epsilon^*(t_s)$ giving DESI match. The naive predictions π^2/t_s (green) and $9.75/t_s$ (blue dotted) are shown for comparison; neither captures the family exactly. The data fix a curve in the (ϵ, t_s) plane along which the cosmogenetic mechanism reproduces DESI; selecting a particular representative requires an additional substrate principle. Reproducible via `code/20_minimum_ts_test.py`.

This degeneracy means that the cosmogenetic data fix the curve, not its individual coordinates: a microscopic theory of the substrate is needed to fix one of the two parameters from first principles, after which the other is predicted by the family. The data alone admit any representative.

5.4 Universal m_0 across the family

The late-time exponent m_0 is found to be *universal* across the (ϵ, t_s) family: numerical scans give $m_0 = 1.57$ to 1.65 across the entire tested range $t_s \in [5, 20]$ and across several alternative source profiles (constant, linear, quadratic, sigmoid; Sec. 5.5). This universality is a *prediction of the substrate dynamics*: once the autocatalysis ceases, the late-time conservative drainage in 3D produces a profile-independent dilution exponent set by the dimension and topology of the lattice, not by the source mechanism. By contrast, m_∞ depends sensitively on where on the family one operates (Sec. 5.3), making the phantom-phase depth a more delicate diagnostic of the cosmogenetic rule.

5.5 Source-profile robustness

The phantom-to-quintessence pattern is not specific to the autocatalytic rule (5): explicit numerical tests (companion code) on five alternative source profiles — constant plateau, linear ramp, quadratic, cubic, and sigmoid — all produce the qualitative phantom-crossing pattern when calibrated to give the same total integrated source energy as the autocatalytic baseline. What changes across profiles is the *depth* of the phantom phase (m_∞) and, to a lesser extent, the late-time exponent m_0. Among the alternatives, the linear profile $S(t) \propto t$ comes closest to DESI ($m_\infty \simeq -3.55$, within $\sim 2\sigma$ of the target); constant, quadratic, cubic and sigmoid profiles deviate more strongly ($|m_\infty - m_\infty^{\rm DESI}| \gtrsim 1$). The autocatalytic rule with the working-point parameters remains empirically the closest match.

5.6 Methodological caveat: extraction sensitivity of m_∞

The numerical value of m_∞ depends measurably on the extraction protocol — the time-grid sampling, the smoothing window for $\langle T^2 \rangle(t)$, and the mask used to define the "phantom plateau". Different reasonable choices can shift the extracted m_∞ by $\Delta m \lesssim 0.7$. The headline value $m_\infty = -3.83 \pm 0.15$ is reproducible *at fixed protocol* across realisations, but the systematic uncertainty from protocol choice should be added in quadrature for cross-comparison with external data. We adopt the protocol of `04_robustness.py` (Savitzky–Golay smoothing, window 21, polynomial order 3, plateau mask $t < 0.7\, t_{\rm peak}$, log-spaced time sampling matching the cosmogenetic timescale) and note this convention explicitly.

5.7 Most stringent null

The most stringent null control remains the *no-shutoff* variant: if the autocatalysis were never to end, no crossing would occur and the framework would fail to explain DESI's evolving signal. Equivalently, the existence of a finite creation epoch t_s is *required* by the data, not just by aesthetic preference. This is a clean falsifier: any future observation showing dark energy continuing to grow (rather than transitioning to quintessence-like dilution) would invalidate the cosmogenetic mechanism as proposed here.

6 Distinguishing predictions

The model differs from ΛCDM and from minimally-coupled quintessence in three testable ways.

(P1) Asymptotic non-de-Sitter future. As $\langle T^2 \rangle \to 0$, the conservative drainage drives all components to zero. This is a "thermal-death" asymptotics, distinct from Λ-driven exponential expansion. Any future observation of a Λ-like late acceleration phase that does *not* asymptote to zero would falsify this prediction.

(P2) Dipolar modulation of H_0. The founding impulse breaks isotropy: an observer at distance r from Ω sees a small dipolar variation of the local cascade arrival time, hence of H_0, scaling as r/L.

(P3) Matter-distribution dipole and quadrupole anomalies. Independent measurements of an H_0 dipole at the $\sim 3\%$ level [1], and an over-large quasar dipole [2] that exceeds the kinematic CMB-dipole prediction, are both consistent with off-centre observer position, although neither is conclusive on its own. Joint detection of a coherent dipole in matter, H_0, and CMB temperature perpendicular to the kinematic dipole direction would constitute a strong positive test of the cosmogenetic mechanism. Persistent failure to find such a signature would falsify it (with the caveat that local and global expansion rates need not coincide).

7 Connection with the cubic substrate of Papers VI–VII

The cosmogenesis cascade fills the Poisson-disk substrate *statistically*; in the late-time regime, the substrate settles toward a homogeneous configuration whose local statistics are expected to match those of the cubic substrate used in Paper III's Floquet sector and Paper VI's oriented sector. The four body-diagonal channels per cube and the spinor S^3 phase space underlying the gauge sector of Paper VI *are conjectured to emerge* as the local geometry of the relaxed substrate after cosmogenesis is complete; an explicit derivation of this Poisson-disk–to–cubic transition is not given here and is left as an open task.

In this reading, the topological ratio $\alpha_G/\alpha_{\mathrm{EM}} = 1/\sqrt{2}$ of Paper VII at the substrate scale, the candidate fixed-point formula for α_{EM} of Paper VI, and the substrate mass scale $m_{\mathrm{lat}} \approx 0.072\, M_P$ are interpreted as properties of the *relaxed* substrate, after cosmogenesis has saturated. Verifying that the relaxation actually produces the required cubic local statistics is open.

8 Limitations and scope

Microscopic origin of ϵ and t_s. The autocatalysis rate $\epsilon = 0.65$ and the creation-epoch duration $t_s = 15$ at the working point are not derived from substrate dynamics. They are jointly degenerate: a one-parameter family of (ϵ, t_s) pairs reproduces the DESI signature (Sec. 5.3). Fixing one of them from a microscopic theory of substrate criticality (for ϵ) or substrate phase transition (for t_s) would predict the other from the family.

$t \leftrightarrow a$ calibration. Identification of simulation time with cosmological scale factor requires a calibration. We adopt the convention that the peak of $\sum T^2$ corresponds to $a^* = 0.69$. A first-principles derivation is the subject of Paper IX.

Pre-recombination physics. We do not address the cosmic microwave background, big-bang nucleosynthesis, or the early radiation-dominated epoch. Quantitatively, applying the linear $t \leftrightarrow a$ calibration $a = a^*(t/t_{\mathrm{peak}})$ with $a^* = 0.69$ to the CMB last-scattering surface ($z \simeq 1100$, $a_{\mathrm{CMB}} \simeq 9 \times 10^{-4}$) maps the CMB epoch to lattice time $t_{\mathrm{CMB}} \simeq 0.020$ – well before the cosmogenetic cascade has built up. At this time the simulation gives $\sum T^2 \simeq 6.5 \times 10^{-3}$, while today ($t_{\mathrm{today}} \simeq 21.7$) it gives $\sum T^2 \simeq 7 \times 10^4$. The simulation's CMB/today ratio is therefore $\sim 9 \times 10^{-8}$, whereas standard cosmology requires $\rho_{\mathrm{CMB}}/\rho_{\mathrm{today}} \sim (1+z)^4 \simeq 1.5 \times 10^{12}$ for the radiation density. The mismatch by ~ 17 orders of magnitude is not a failure of the framework but a precise statement of its scope: the cosmogenetic mechanism of this paper describes only the late-time ($z \lesssim 1$) phenomenology, not the radiation- dominated era. Reproducing the CMB requires either an earlier cosmogenetic event that loads the substrate with high radiative content prior to the cascade considered here, or a different non-perturbative DDD phase governing the radiation era. We propose this as the subject of Paper X (Pre-recombination cosmogenesis). This back-of-the-envelope mismatch is included not as a cosmological claim, but as a guardrail against overextending the late-time calibration.

Absolute Hubble constant. We do not predict H_0 in absolute units; only its angular variation.

9 Open questions

Derivation of ϵ from substrate criticality. Numerical experiments suggest that ϵ is *not* simply set by the discrete Laplacian eigenvalue ($\epsilon \propto \alpha/\bar{D}$ is excluded by direct test); the measured $\epsilon^*(t_s)$ family does not collapse onto a simple geometric ratio. Identifying the substrate principle that fixes ϵ along the family is the central theoretical task left open.

Microscopic origin of the creation epoch. Why does the substrate admit a finite "creation epoch" $[0, t_s]$ during which non-conservation is active, followed by strict conservation? Candidate mechanisms include capacity saturation, phase transition, or slow decay of ϵ with cumulative cascade content.

Universal m_0 from substrate dimension. The empirical universality $m_0 \simeq +1.6$ across the (ϵ, t_s) family and across all eight non-trivial source profiles tested suggests a derivation from the conservative late-time drainage of a 3D Poisson-disk lattice. A direct analytic calculation is open.

Observers' positions. Constraining r/L from the joint H_0 dipole, matter dipole, and CMB dipole directions is the target of Paper XI.

10 Code and data availability

All code is in the `code/` subdirectory:

- `01_simulation.py` – core cascade simulation (variant A)
- `04_robustness.py` – 10-realisation ensemble (variant A)
- `07_phase_heatmap.py` – chronological heatmap (Fig. 1)
- `09_radial_profile.py` – isotropic radial profile
- `10_isotropic_snapshots.py` – isotropic 2D snapshots
- `11_source_profile_scan.py` – five power-law profiles
- `14_autosource_test.py` – variants (D), (E), (F)
- `16_autosource_ensemble.py` – **headline 10-realisation ensemble for variant (E)**
- `20_minimum_ts_test.py` – (ϵ, t_s) family

- `21_uniform_global_test.py` – variant (G) trivial null
- `22_abis_test.py` – variant (A.bis) self-tuned amplitude
- `23_delta_init_test.py` – variant (H) delta-init

11 Outlook

1. Paper IX (effective cosmology) will treat the *dilution–expansion equivalence* systematically.
2. Paper XI (*Our Position in the Cosmogenesis*) will quantitatively analyse off-centre observers and the resulting H_0 dipole, bulk flow toward the Great Attractor, and four-way directional alignment of late-time anisotropies.
3. A first-principles derivation of ϵ from substrate criticality is the principal theoretical open task.

The principal contribution of this paper is to show that the DESI DR2 evolving dark-energy signal emerges naturally from the conservative drainage rule of Paper I under a finite-epoch local autocatalysis rule, with no externally-imposed exponential time profile.

12 Conclusion

If the late-time cosmological dynamics is the macroscopic signature of a discrete substrate just now finishing the cascade that filled it from a single founding event, then the DESI DR2 phantom crossing is the inevitable transition between filling and dilution. The adoption of a local autocatalysis rule replaces the externally imposed exponential source by a substrate-level rate; the resulting mechanism reproduces the DESI signature and reveals a one-parameter (ϵ, t_s) degeneracy along which a microscopic theory of the substrate could fix the cosmogenetic epoch.

The result should not be read as a first-principles cosmology: the autocatalysis rate ϵ, the creation-epoch duration t_s, the FRW distance interpretation, and the radiation-dominated era all remain open. Its contribution is narrower but concrete: a conservative drainage cascade with finite-epoch self-amplification produces a phantom-to-quintessence transition with the right qualitative shape and quantitative parameters at $z \lesssim 1$, with the late-time exponent m_0 emerging as a universal substrate-geometric prediction.

References

[1] P. Boubel, M. Colless, K. Said, and L. Staveley-Smith. Testing anisotropic Hubble expansion. *arXiv e-prints*, 2024.

[2] N. J. Secrest, S. von Hausegger, M. Rameez, R. Mohayaee, S. Sarkar, and J. Colin. A Test of the Cosmological Principle with Quasars. *Astrophys. J. Lett.*, 908:L51, 2021.

[3] DESI Collaboration, M. Abdul Karim, et al. DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints. *arXiv e-prints*, 2025.

[4] K. Lodha et al. Extended Dark Energy analysis using DESI DR2 BAO measurements. *Phys. Rev. D*, 112:083511, 2025.

[5] I. D. Gialamas, G. Hütsi, et al. Quintessence and phantoms in light of DESI 2025. *arXiv e-prints*, 2025.

[6] H. Adam, M. Hertzberg, D. Jiménez-Aguilar, and I. Khan. Comparing Minimal and Non-Minimal Quintessence Models to 2025 DESI Data. *arXiv e-prints*, 2025.